

Task 5: AI Career Roleplay

Name

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Task Title

AI-Based Student Performance Prediction System

Role

AI Engineer

Objective

The objective of this project is to design an Artificial Intelligence system that predicts student academic performance and identifies students who may require additional support. The system helps educational institutions improve student outcomes through data-driven decision-making.

Introduction

Artificial Intelligence (AI) is transforming education by enabling institutions to analyze large amounts of student data and make intelligent predictions. By studying factors such as attendance, study hours, and academic scores, AI can help identify students who are at risk of poor performance and provide recommendations for improvement.

Problem Statement

Teachers often find it difficult to monitor every student's progress individually. As a result, students facing academic challenges may not receive support early enough. An AI-based prediction system can analyze student data and provide early warnings, allowing teachers to take corrective action before final examinations.

Data Collection

The AI system collects the following information:

- Student Attendance Percentage
- Study Hours per Day
- Assignment Completion Status
- Internal Assessment Marks
- Class Participation

- Previous Examination Scores

This data is used to understand student learning patterns and predict future performance.

Data Processing

The collected data is processed through the following steps:

1. Data Collection
2. Data Cleaning
3. Feature Selection
4. Model Training
5. Prediction Generation
6. Recommendation Generation

These steps ensure that the AI system produces reliable predictions.

Sample Dataset

Attendance	Study Hours	Marks
95	5	88
80	4	75
60	2	45
90	6	92
50	1	5

```
[1] import pandas as pd

data = {
    "Attendance": [95, 80, 60, 90, 50],
    "Study_Hours": [5, 4, 2, 6, 1],
    "Marks": [88, 75, 45, 92, 35]
}

df = pd.DataFrame(data)

print(df)
```

...	Attendance	Study_Hours	Marks
0	95	5	88
1	80	4	75
2	60	2	45
3	90	6	92
4	50	1	35

Prediction Logic

The system analyzes student marks and classifies students into categories.

- Marks greater than or equal to 50 → Pass

- Marks below 50 → Needs Improvement

Example Output:

Pass

Pass

Needs Improvement

Pass

Needs Improvement



```
for mark in df["Marks"]:  
    if mark >= 50:  
        print("Pass")  
    else:  
        print("Needs Improvement")
```

... Pass
Pass
Needs Improvement
Pass
Needs Improvement

AI Workflow

Input Student Data



Data Cleaning



Feature Selection



AI Model Training



Prediction



Student Recommendation

Benefits of the System

- Early identification of struggling students
- Improved academic performance
- Better decision-making for teachers
- Personalized learning recommendations
- Efficient use of educational resources

Technologies Used

- Python
- Pandas
- Google Colab
- Artificial Intelligence Concepts
- GitHub

Learning Outcomes

Through this project, I learned:

- How AI systems solve real-world problems
- The importance of data collection and preprocessing
- Basic prediction logic
- Data analysis using Python
- AI workflow design and documentation

Conclusion

The AI-Based Student Performance Prediction System demonstrates how Artificial Intelligence can be used to improve educational outcomes. By analyzing student data and predicting performance, educational institutions can provide timely support and help students achieve better academic results. This project enhanced my understanding of AI applications, data-driven decision-making, and predictive analytics.